

Technical Architecture

The proposed architecture is a ServiceNow cloud-platform workflow architecture. The requester interacts with the Service Catalog, the request is represented by a Requested Item record, and Flow Designer / Workflow Studio automates approval, catalog-task creation and email notification.

Table-1: Components & Technologies

Table-2: Application Characteristics

Architecture Flow

1. Requester opens the Car Rental Request catalog item.
2. Requester enters rental variables and submits the request.
3. ServiceNow creates the Requested Item (RITM).
4. Flow Designer asks for approval.
5. If the approval condition is true, the flow creates a Catalog Task.
6. The Catalog Task is assigned to the configured ITSM Engineering fulfillment group.
7. The flow sends the configured email notification.
8. Administrators and fulfillment users can review the request and task records in ServiceNow.

References section follows the structure of the supplied template. The implementation itself is based on the ServiceNow platform used for this project.

Date	_____ 23 September 2026 _____		
Team ID	_____		
Project Name	Car Rental Request Automation in ServiceNow Using Service Catalog and Workflows		
Maximum Marks	4 Marks		
S.No	Component	Description	Technology
1	User Interface	How the requester interacts with the car rental request.	ServiceNow Service Catalog / Service Portal
2	Application Logic-1	Captures the request and manages catalog-request processing.	ServiceNow Service Catalog
3	Application Logic-2	Automates approval and condition-based flow execution.	ServiceNow Flow Designer / Workflow Studio
4	Application Logic-3	Creates the fulfillment task after approval.	ServiceNow Catalog Task / Flow Designer
5	Database	Stores catalog requests, Requested Items, approvals, users and catalog tasks.	ServiceNow platform database
6	Cloud Database	Cloud-hosted application data storage provided by the ServiceNow platform.	ServiceNow cloud platform
7	File Storage	No separate file-storage service is required for the core workflow.	Not applicable
8	External API-1	No external API is required by the implemented workflow.	Not applicable
9	External API-2	No second external API is required by the implemented workflow.	Not applicable
10	Machine Learning Model	No machine-learning model is used in this project.	Not applicable

S.No	Component	Description	Technology
11	Infrastructure (Server / Cloud)	Application and workflow run on the ServiceNow instance.	ServiceNow cloud platform

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is required for the core implementation.	Not applicable; ServiceNow platform
2	Security Implementations	Access to records and configuration is controlled through ServiceNow authentication, roles and access controls.	ServiceNow roles / ACL-based access control
3	Scalable Architecture	The solution separates request capture, approval and fulfillment automation into platform components that can be extended with additional catalog items, tasks and flow logic.	ServiceNow Service Catalog + Flow Designer
4	Availability	The application is accessed through the ServiceNow cloud instance rather than a locally hosted application server.	ServiceNow cloud platform
5	Performance	The flow uses event-driven automation: a submitted catalog request triggers approval and subsequent fulfillment actions, avoiding manual processing steps.	ServiceNow Flow Designer

Level 0 / High-Level Data Flow

